

ORIGINAL RESEARCH

ECG Signatures and Long-Term Ischemic Stroke Risk



A Deep Learning Analysis of 200,000 Patients

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ABSTRACT

BACKGROUND Scalable risk stratification for ischemic stroke remains an unmet need.

OBJECTIVES In this study, the authors sought to assess whether deep learning of 12-lead electrocardiograms (ECGs) can estimate longitudinal ischemic stroke risk and quantify the extent to which risk signals reflect plausible mechanisms (eg, atrial cardiopathy).

METHODS We trained a convolutional neural network to estimate the 10-year risk of incident ischemic stroke with the use of 12-lead ECG among patients receiving longitudinal care at Massachusetts General Hospital (MGH). Neural network-derived stroke probabilities, age, and sex were integrated into a Cox proportional hazards model ("ECG2Stroke"). Within an MGH test set ("MGH Test"), as well as independent samples from Brigham and Women's Hospital (BWH) and Beth Israel Deaconess Medical Center (BIDMC), we assessed model discrimination (area under the curve [AUC]) and calibration (integrated calibration index [ICI]). ECG2Stroke was compared with the revised Framingham Stroke Risk Profile (FSRP). Saliency mapping, associations with clinical factors and structured ECG features, and performance across stroke subtypes were assessed.

RESULTS ECG2Stroke was developed in 101,496 individuals from MGH (age 57 ± 16 years, 48% women), and evaluated in MGH Test ($n = 4,771$; age 57 ± 16 years, 49% women), BWH ($n = 68,884$; age 57 ± 16 years, 55% women), and BIDMC ($n = 29,882$; age 56 ± 17 years, 54% women). At 10 years, there were 346 stroke events in MGH Test, 3,209 in BWH, and 1,236 in BIDMC. ECG2Stroke demonstrated moderate discrimination of incident stroke (10-year AUCs: MGH Test, 0.795; BWH, 0.774; BIDMC, 0.772) and low calibration error (ICIs: MGH Test, 0.030; BWH, 0.005; BIDMC, 0.026). In patients with available data, 10-year AUC for ECG2Stroke was similar to FSRP (MGH/BWH Test: ECG2Stroke, 0.791; FSRP, 0.779; BIDMC: ECG2Stroke, 0.745; FSRP, 0.728). Stratification persisted across subgroups, including patients with and without atrial fibrillation. Saliency maps highlighted the ECG P-wave, and risk estimates correlated with structured P-wave indices. ECG2Stroke was strongly associated with cardioembolic stroke (cause-specific HR: 2.17 per 1-SD of logit-transformed probability; 95% CI: 1.64–2.87) but not noncardioembolic stroke.

CONCLUSIONS ECG-based artificial intelligence (AI) can predict 10-year ischemic stroke with performance similar to a validated clinical score, possibly by encoding markers of abnormal atrial substrate linked to cardioembolism. AI-enabled ECG analysis may enable efficient prioritization for stroke prevention. (JACC. 2026;88:205–223) © 2026 by the American College of Cardiology Foundation.

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ABBREVIATIONS AND ACRONYMS

AF	= atrial fibrillation
AP	= average precision
ASCVD	= atherosclerotic cardiovascular disease
CE	= cardioembolic
ECG	= electrocardiography
EHR	= electronic health record
FSRP	= Framingham Stroke Risk Profile
HF	= heart failure
ICI	= integrated calibration index
LP	= linear predictor
MI	= myocardial infarction
nAP	= normalized average precision, the ratio of the model's AP to the baseline event rate
OAC	= oral anticoagulation

Stroke remains a leading cause of death and disability worldwide.¹ Although age-adjusted rates of stroke have declined, the absolute number of strokes occurring annually continues to increase. Recurrent strokes account for only one-fourth of stroke events, and ischemic stroke is the most common type, highlighting a gap in primary prevention.²

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It is estimated that more than one-half of stroke events could be prevented by implementing established preventive measures (eg, atherosclerotic risk factor modification).^{2,3} Although additional ischemic stroke risk factors, such as atrial fibrillation (AF) or carotid artery stenosis, are actionable in high risk groups once detected, there are currently no clearly effective methods for mass screening.³⁻⁶ Thus, population-level approaches to accurately identify individuals at

elevated risk for stroke who should be prioritized for intensive prevention remain a critical unmet need.

Clinical stroke risk prediction tools such as the Framingham Stroke Risk Profile⁷ (FSRP) are capable of stratifying 10-year risk of stroke, but such models are limited by the need for multiple clinical variables and requirement for cumbersome calculation, limiting their widespread use. In contrast, the 12-lead electrocardiogram (ECG) represents a low-cost ubiquitous diagnostic test well suited for population-level risk estimation. Although various ECG characteristics have been linked to stroke risk, existing efforts have largely relied on overtly abnormal ECG findings, specific waveform parameters not routinely quantified, or prediction of known intermediary risk factors such as AF.⁸⁻¹¹ In contrast, we have previously used artificial intelligence (AI)-enabled analysis of raw ECG waveform data to directly predict future incidence of important cardiovascular outcomes¹²⁻¹⁴; whether such an approach may accurately estimate risk of future stroke remains unknown.

Furthermore, AI-enabled ECG models have recently shown overlapping associations among

some left ventricular and left-side valvular phenotypes; the extent to which such overlap reflects a general tendency to encode overall cardiovascular risk remains unknown.¹⁵ In this context, systematically characterizing ECG and clinical correlates of an ECG-AI model for stroke may provide insight into the extent to which ECG-based AI models may encode risk signals specific to underlying physiologic substrate (eg, atrial cardiopathy), which may in turn drive associations between ECG waveform analysis and future ischemic stroke.

In this study, we develop ECG2Stroke, a model that leverages deep learning neural networks to predict 10-year ischemic stroke risk directly from baseline 12-lead ECGs obtained in the context of routine ambulatory care from >100,000 patients and validate the model in multiple independent datasets totaling >97,000 patients. We benchmark its performance against the FSRP and systematically examine its clinical and ECG associations to interrogate potential mechanisms linking ECGs to ischemic stroke.

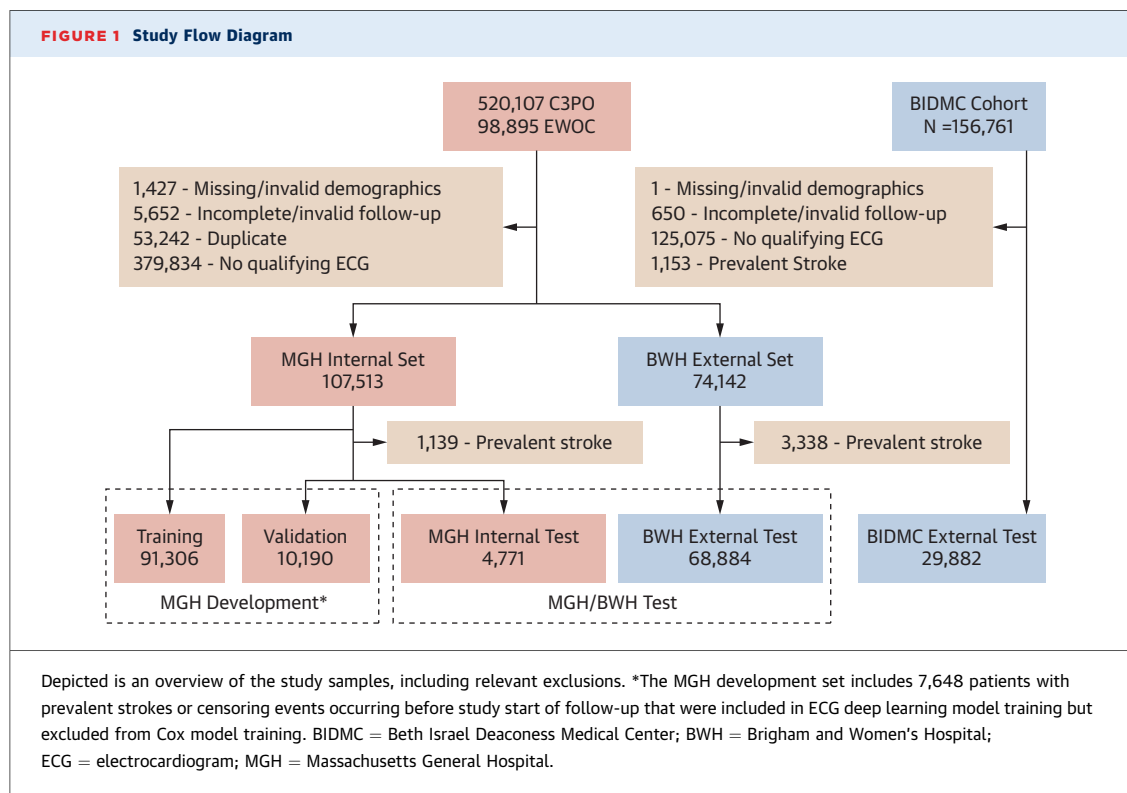
METHODS

The use of patient data for this study was approved by the Mass General Brigham (MGB) and Beth Israel Deaconess Medical Center (BIDMC) institutional review boards.

STUDY POPULATION. ECG2Stroke was developed with the use of data from 2 previously described patient cohorts, the Community Care Cohort Project (C3PO)¹⁶ and the Enterprise Warehouse of Cardiology (EWOC),¹⁷ representing ~520,000 patients receiving longitudinal primary care and ~99,000 patients receiving longitudinal cardiology care, respectively, within the MGB multi-institutional health care system in the New England region of the U.S. Briefly, patients aged 18 to 90 years receiving longitudinal primary or ambulatory cardiology care within 1 of 7 MGB network hospitals sharing a unified electronic health record (EHR) from 2000 to 2019 were included. As described in the original publications, patients needed to have at least 2 qualifying visits 1 to 3 years apart within primary care (C3PO) or

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).



ambulatory cardiology (EWOC) clinics for inclusion, resulting in a longitudinal data structure previously shown to demonstrate favorable data density with less selection bias than traditional EHR sampling methods.

The deep learning neural network was trained on patients with ≥ 1 ECG at Massachusetts General Hospital (MGH) within 3 years before the start of cohort follow-up (defined as the second of the first qualifying pair of visits required for C3PO or EWOC inclusion). A validation set of patients from MGH without previous history of stroke was used for hyperparameter tuning and model selection. A neural network-derived model was then tested on independent test sets of stroke-free individuals with eligible ECGs from MGH (internal test set) and Brigham and Women's Hospital (BWH) (external test set). Patients with ECGs from both hospitals (4,600 individuals) retained only their MGH-performed ECGs and were included only in the MGH cohort, without inclusion in the BWH cohort. There was therefore no overlap at the individual or ECG level between the training, validation, and test sets. We performed additional external validation among patients receiving longitudinal primary care at Beth Israel Deaconess Medical Center (BIDMC), an independent health care system, and with no history of stroke (external test set) (Figure 1).

Although they are all located within the larger Boston region, MGH, BWH, and BIDMC have distinct patterns of community-based and outside referrals, case mix, and health care utilization, as outlined in the 2022 Massachusetts Center for Health Information and Analysis report.¹⁸ BIDMC is further distinct as part of an entirely separate health system. For example, top community discharges in 2020 were as follows: BWH: Jamaica Plain, Brookline, Roslindale; MGH: Charlestown, Chelsea, Revere; BIDMC: Brookline, Jamaica Plain, Roxbury. Top discharge diagnoses by diagnosis-related group were as follows: BWH: craniotomy, cardiovascular procedures without acute myocardial infarction, major small and large bowel procedures; MGH: moderately extensive procedure unrelated to principal diagnosis, craniotomy, major respiratory infection and inflammation, percutaneous cardiovascular procedures without acute myocardial infarction; BIDMC: normal neonate birth, vaginal delivery, cesarean section, cardiovascular procedures without acute myocardial infarction.

ECG ACQUISITION AND NEURAL NETWORK TRAINING. The input to the deep learning neural network is a single 12-lead ECG, representing a 10-second time-series sampled at 500 Hz for each of the 12 leads. To create a uniform input tensor of

dimension $5,000 \times 12$, ECGs with lower sampling rates (a minority) were linearly up-sampled to 500 Hz, and those with shorter durations were zero-padded to 5,000 measurements. Raw waveform data and tabular metadata (including date, time, machine type, sampling frequency, and automated and physician reads) were extracted from all ECGs. All ECGs had been performed for clinical purposes. No ECGs were excluded from the training or test sets on the basis of particular clinical findings.

The neural network used a loss function that accounted for both time to outcome (ie, ischemic stroke) and missingness due to censoring (defined as the earliest of death or last encounter). This loss function was previously used to predict 5-year AF-free survival.¹² Similarly to previous work, we also trained the network to perform related secondary tasks (prediction of 10-year overall survival, age estimation, sex classification, and identification of AF on the ECG). This multitask approach improved stroke prediction performance compared with other model-building approaches considered during model derivation ([Supplemental Table 1](#)). Further details are provided in the [Supplemental Methods](#).

The neural network was trained with the use of all 12-lead ECGs in the MGH training dataset performed within 3 years before the start of follow-up. Because ECGs from individuals with prevalent disease are likely to contain rich training signals, patients with prior stroke were included in the training set, consistently with prior studies.¹² However, individuals with prevalent stroke were excluded from all validation and test sets.

The neural network is a deep 1-dimensional convolutional network. The architecture is summarized in [Supplemental Figure 1](#). Briefly, the network consists of multiple densely connected convolutional blocks followed by several fully connected layers. These layers combine information from the entire 12-lead ECG to produce a longitudinal estimate of stroke-free survival. Learning curves are shown in [Supplemental Figure 2](#), and the performance of the neural network on secondary tasks is shown in [Supplemental Figure 3](#).

CLINICAL FACTORS. Baseline age, sex, and race were obtained from the EHR. Race/ethnicity was classified as White, Black, Hispanic/Latino, Asian or Pacific Islander, or other/unknown. Tobacco use was categorized as present or absent. The presence of hypertension, hyperlipidemia, diabetes, chronic kidney disease, heart failure, valvular disease, and myocardial infarction was ascertained according to previously published sets of diagnostic codes and

medications (eg, antidiabetics), having a reported positive predictive value (PPV) $\geq 85\%$ in MGH and BWH.^{19,20} AF was defined using a previously published algorithm comprising diagnostic and procedural codes and ECG reports (PPV 92%).²¹ Antihypertensive use was determined according to medication lists.¹⁹ The presence of obesity was similarly defined with the use of diagnosis codes. Use of oral anticoagulants and statin drugs was determined with the use of medication lists (for oral anticoagulants, outpatient only). Blood pressure values, low-density lipoprotein, and laboratory values were obtained from the EHR. Analogous definitions were applied in BIDMC. Disease definitions are listed in detail in [Supplemental Table 2](#).

OUTCOMES. The primary outcome was incident ischemic stroke, defined according to previously published diagnosis code sets and having PPV $\geq 85\%$ in MGH and BWH and strong pairwise agreement between manual reviewers (Cohen's $\kappa = 0.8$), and modified to remove codes specific for transient ischemic attack (TIA).¹⁹ The entirety of the EHR (regardless of treating site) was used for outcome ascertainment. A similar definition was used in BIDMC ([Supplemental Table 2](#)).

STATISTICAL ANALYSIS. Our primary objective was prediction of incident ischemic stroke at 10 years in the MGH, BWH, and BIDMC test sets. Shorter time horizons at 1 and 5 years were also assessed. During model evaluation, patients with stroke prevalent at the start of follow-up were excluded and only the most recent ECG before start of follow-up was used for each individual. To avoid misclassification of prevalent strokes as incident due to delayed coding, a 30-day blanking period was applied such that person-time began 30 days after the start of cohort follow-up. Events occurring during the 30-day blanking period were considered prevalent. To correct for miscalibration introduced by inclusion of patients with prevalent stroke in the neural network training process, we fitted Cox proportional hazards models including neural network-predicted event probabilities alone or with age and sex as covariates for 1-, 5-, and 10-year horizons in the subset of the combined training and validation sets (MGH development set) with prevalent stroke excluded. Predicted event probabilities were logit transformed to achieve an approximately linear relationship with the log hazard.²² Given a low death rate when considering person-time before stroke or within 365 days of the last encounter (8.8% MGH development set), death was treated as a censoring event. Sensitivity analyses were subsequently performed with the use of Fine-

Gray models and the cumulative incidence function with death as a competing risk. Given our intention to assess the feasibility of population-level stroke risk stratification with the use of a single 12-lead ECG, and the universal availability of age and sex in this context, the Cox model combining neural network-predicted stroke probabilities with age and sex was selected as our primary model, dubbed ECG2Stroke. A statistically significant interaction was observed during training between neural network-predicted probabilities and male sex (HR: 0.91; 95% CI: 0.85-0.98), but the interaction was not included in the final model due to only marginal improvement in Akaike Information Criterion. There was no significant interaction with age. ECG2Stroke model parameters are provided in [Supplemental Table 3A](#). Although it was used as a secondary training target for the neural network, presence of AF in the diagnostic statement of the qualifying ECG as an additional covariate was not a significant predictor and caused minimal change in remaining ECG2Stroke regression coefficients, and therefore was not included in the final model.

To compare the performance of ECG2Stroke with the revised FSRP,⁷ we further restricted the MGH development set to patients with complete data (22%) for calculation of the FSRP and fitted analogous Cox proportional hazard models, but including as covariates predicted event probabilities from ECG2Stroke alone, FSRP alone, and both ECG2Stroke and FSRP. Additional baseline models based on ECG alone or age and sex were also fitted in the same population. No significant interaction was observed between neural network probabilities with either age, sex, or FSRP. Model fit parameters for ECG2Stroke in the FSRP complete data subset are provided in [Supplemental Table 3B](#).

Discrimination and calibration were evaluated in the test sets by calculating the time-dependent area under receiver operating characteristic curve (AUC),²³ time-dependent average precision (AP),²⁴ and integrated calibration index²⁵ (ICI) fit with the use of restricted cubic splines. AUC and AP estimates were calculated with the use of inverse probability of censoring weights. AP was normalized to the baseline event rate (AP of a noninformative model) and reported as nAP. The *P* values for AUC and AP comparisons and CIs were obtained by bootstrap. The same evaluation metrics were assessed in subgroups of age, sex, race, and clinical risk factors in external test sets (BWH and BIDMC test sets). Incremental model contribution was quantified by means of variance decomposition in Cox models, reporting

partial R^2 derived from likelihood-ratio testing to estimate the proportion of explained variation.²⁶ We explored stroke-free survival in high and low risk strata predicted by ECG2Stroke vs FSRP models with the use of risk thresholds (predicted 10-year risk <5% for low risk, >20% for high risk) derived from standard atherosclerotic cardiovascular disease (ASCVD) prevention guidelines.²⁷ Survival differences were compared by means of log-rank testing. Reclassification at 10-year horizon was quantified with time-dependent net reclassification improvement (NRI) using the inverse probability weighting method²⁸ and CIs obtained by bootstrap. Decision curves were constructed across a range (0.01-0.45) of predicted 10-year risk thresholds.²⁹ We evaluated the decision curves at selected thresholds of 10-year risk (0.05, 0.075, 0.10, 0.20) to report net benefit and net interventions avoided, and quantified the corresponding classification performance (sensitivity, specificity, PPV, and negative predictive value [NPV]) implied by the decision rules. We computed the linear predictors of ECG2Stroke and FSRP-derived Cox in the test sets and assessed Pearson correlation.

SECONDARY ANALYSES AND SALIENCY. Additional analyses examined the performance of ECG2Stroke across strata of prevalent AF (MGH, BWH, and BIDMC test sets) and baseline oral anticoagulation (OAC) use (MGH and BWH only). Among patients with prevalent AF, stroke-free survival and reclassification was examined for high and low risk strata defined with the use of guideline-based thresholds according to the CHA₂DS₂-VASc³⁰ score (low risk: <1 for men and <2 for women; high risk: >1 for men and >2 for women) vs high and low risk strata defined according to the ECG2Stroke thresholds listed above. To ensure adequate sample size, the MGH and BWH test sets were combined into the MGH/BWH test set for analyses restricted to specific subgroups such as the FSRP complete data subset or prevalent AF subset. Decision curves for CHA₂DS₂-VASc mapped integer scores to corresponding Kaplan-Meier estimates. We further modeled AF status as a time-dependent variable alone or jointly with ECG2Stroke to assess marginal contributions to risk prediction by inspecting their interaction, coefficient attenuation, and partial R^2 .

To assess ECG features driving the neural network's predictions, we plotted median ECG waveforms and produced saliency maps using a random sample of 7,080 individuals from the BWH test set. These saliency maps depicted areas of the ECG where changes in voltage had the greatest influence on the

TABLE 1 Baseline Characteristics

	MGH Dev. Set* (n = 101,496)	MGH Test Set (n = 4,771)	BWH Test Set (n = 68,884)	BIDMC Test Set (n = 29,882)
Age, y	57.1 ± 16.4	57.0 ± 16.1	56.6 ± 15.8	56.3 ± 16.6
Female	48,925 (48.2)	2,342 (49.1)	37,914 (55.0)	16,252 (54.4)
Race/ethnicity				
White	82,066 (80.9)	3,873 (81.2)	50,207 (72.9)	15,795 (52.9)
Black	5,534 (5.5)	270 (5.7)	7,541 (10.9)	4,705 (15.7)
Hispanic or Latino	4,036 (4.0)	182 (3.8)	4,446 (6.5)	1,733 (5.8)
Asian or Pacific Islander	3,792 (3.7)	152 (3.2)	1,668 (2.4)	857 (2.9)
Other/unknown	6,068 (6.0)	294 (6.2)	5,022 (7.3)	6,792 (22.7)
Current smoker	5,501 (5.4)	262 (5.5)	5,560 (8.1)	1,899 (6.4)
Diabetes	19,194 (18.9)	876 (18.4)	12,449 (18.1)	8,453 (28.3)
Hyperlipidemia	58,896 (58.0)	2,775 (58.2)	31,473 (45.7)	17,356 (58.1)
Chronic kidney disease	9,896 (9.8)	443 (9.3)	6,568 (9.5)	1,983 (6.6)
Hypertension	73,133 (72.1)	3,411 (71.5)	49,496 (71.9)	17,927 (60.0)
Systolic blood pressure, mm Hg	126.9 ± 17.7	126.9 ± 17.9	127.9 ± 17.9	127.5 ± 17.8
Antihypertensive medication	58,505 (57.6)	2,731 (57.2)	43,852 (63.7)	11,308 (37.8)
Atrial fibrillation	16,821 (16.6)	715 (15.0)	11,060 (16.1)	2,817 (9.4)
Heart failure	16,894 (16.6)	738 (15.5)	10,597 (15.4)	3,103 (10.4)
Valvular disease	9,671 (9.5)	400 (8.4)	7,117 (10.3)	927 (3.1)
Vascular disease	42,510 (41.9)	1,853 (38.8)	26,123 (37.9)	8,490 (28.4)
Prevalent stroke	5,464 (5.4)	0 (0.0)	0 (0.0)	0 (0.0)
Incident stroke†	4,591 (4.5)	346 (7.3)	3,209 (4.7)	1,236 (4.1)
Follow-up duration, y	7.0 (2.9-12.9)	7.5 (3.2-13.2)	7.0 (2.9-12.0)	10.8 (5.2-15.3)

Values are mean ± SD, n (%), or median (Q1-Q3). *MGH development here reflects the combined internal training and validation sets used to develop the ECG Neural Network model. It includes 7,648 patients with prevalent strokes or censoring events occurring before study start of follow-up that were included in ECG deep learning model training but excluded from Cox model training. †New stroke diagnosis within 10 years.

BIDMC = Beth Israel Deaconess Medical Center; BWH = Brigham and Women's Hospital; MGH = Massachusetts General Hospital.

model's predicted risk estimates. Further details of saliency map and median waveform analyses are described in the [Supplemental Methods](#). To further characterize the neural network, we calculated Pearson correlation of its output probabilities (logit transformed) to known P-wave indices (eg, PR interval, P-wave duration).

We assessed ECG2Stroke and FSRP outcome specificity by examining discrimination (AUC) and calibration (ICI) performance for incident heart failure (HF), myocardial infarction (MI), and death across all test sets. In a small subset of patients with incident strokes with manual physician-adjudicated stroke subtype information, cause-specific Cox models were used to estimate associations with cardioembolic (CE) vs non-CE stroke subtypes, with heterogeneity assessed according to the Lunn-McNeil method.³¹ Further details of the stroke subtyping labels are provided in the [Supplemental Methods](#). We characterized ECG2Stroke risk patterns by assessing associations between its output probabilities (logit transformed) and the presence or absence of baseline clinical stroke risk factors using age- and sex-adjusted linear regression models. Analyses were

performed with the use of Python v3.13³² and R v4.5.³³

RESULTS

TRAINING AND VALIDATION SAMPLES. A total of 107,513 patients from MGH were further subdivided into training (n = 91,306) and validation (n = 10,190) sets used for model development, as well as an internal test (n = 4,771) set. The training set included 299,640 12-lead ECGs (median 6 ECGs per individual). The MGH training and validation sets were collapsed into a single MGH development set (n = 101,496) for parameterization of Cox models. Separate external test sets were constructed from patients with ECGs performed at BWH (n = 68,884) and BIDMC (n = 29,882). **Figure 1** depicts a study flow diagram. The proportion of patients with an incident stroke event within 10 years was 4.5% in the MGH development set over a median follow-up duration of 7.0 years, 7.3% over 7.5 years in the MGH internal test set, 4.7% over 7.0 years in the BWH external test set, and 4.1% over 10.8 years in the BIDMC external test set.

TABLE 2 ECG2Stroke Performance for 10-Year Incident Stroke Prediction in Test Sets

	HR (per SD)	AUC	AP	nAP	ICI
MGH test set (n = 4,771; event rate = 0.104)					
Age + sex	2.609 (2.267-3.002)	0.781 (0.752-0.808)	0.181 (0.159-0.209) ^a	1.740 (1.530-2.010)	0.033
ECG only	2.483 (2.197-2.807)	0.767 (0.740-0.793) ^a	0.202 (0.173-0.239)	1.940 (1.670-2.290)	0.029
ECG2Stroke	2.838 (2.482-3.245)	0.795 (0.768-0.820)	0.205 (0.179-0.237)	1.970 (1.720-2.280)	0.030
BWH test set (n = 68,884; event rate = 0.072)					
Age + sex	2.262 (2.165-2.365)	0.736 (0.726-0.746) ^a	0.112 (0.107-0.117) ^a	1.560 (1.490-1.640)	0.004
ECG only	2.370 (2.277-2.467)	0.755 (0.745-0.764) ^a	0.127 (0.120-0.134)	1.770 (1.680-1.870)	0.008
ECG2Stroke	2.648 (2.535-2.765)	0.774 (0.765-0.783)	0.127 (0.121-0.133)	1.760 (1.680-1.860)	0.005
BIDMC test set (n = 29,882; event rate = 0.053)					
Age + sex	2.372 (2.212-2.545)	0.734 (0.720-0.749) ^a	0.092 (0.085-0.100) ^a	1.740 (1.610-1.890)	0.024
ECG only	2.558 (2.392-2.735)	0.756 (0.743-0.769) ^a	0.103 (0.095-0.112)	1.940 (1.790-2.120)	0.025
ECG2Stroke	2.803 (2.609-3.012)	0.772 (0.758-0.785)	0.104 (0.096-0.112)	1.950 (1.810-2.120)	0.026

Ranges in parentheses are 95% CIs. ^aP < 0.05 compared with ECG2Stroke.
AP = average precision (event rate = baseline event rate, corresponding to the AP of a noninformative model); ICI = integrated calibration index; nAP = normalized average precision is the ratio of the model's AP to the baseline event rate; other abbreviations as in Table 1.

Baseline sample characteristics are summarized in Table 1.

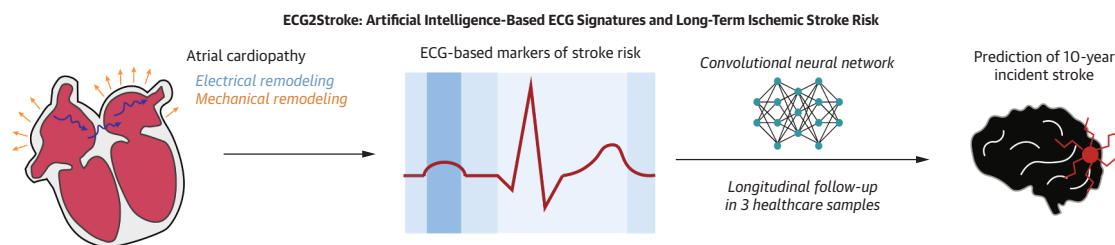
MODEL PERFORMANCE IN TEST SETS. ECG2Stroke demonstrated strong univariable association with 10-year incident stroke in the test sets (HR per SD: MGH, 2.838 [95% CI: 2.482-3.245]; BWH, 2.648 [95% CI: 2.535-2.765]; BIDMC, 2.803 [95% CI: 2.609-3.012]) and exhibited moderate discrimination of incident stroke (10-year AUC: MGH, 0.795 [95% CI: 0.768-0.820]; BWH, 0.774 [95% CI: 0.765-0.783]; BIDMC, 0.772 [95% CI: 0.758-0.785]) (10-year nAP: MGH, 1.970 [95% CI: 1.720-2.280]; BWH, 1.760 [95% CI: 1.680-1.860]; BIDMC, 1.950 [95% CI: 1.810-2.120]) (Table 2). Estimates were generally well calibrated (10-year horizon ICI: MGH, 0.030; BWH, 0.005; BIDMC, 0.026) (Table 2, Central Illustration), and stratified longitudinal stroke risk (Central Illustration). AUC values were largely similar at shorter time horizons (5-year horizon range: 0.753-0.765; 1-year horizon range: 0.701-0.749) (Supplemental Tables 4A and 4B), while nAP improved with shorter time horizons (10-year horizon range: 1.760-1.970; 5-year horizon range: 2.110-2.450; 1-year horizon range: 2.470-3.190) (Table 2, Supplemental Tables 4A and 4B).

ECG2Stroke demonstrated consistently higher average precision compared with a model based on age and sex alone and generally favorable AUC compared with models based on ECG alone or age and sex alone (Table 2, Supplemental Table 4A). Variance decomposition confirmed that the ECG alone model explained a significant proportion of additional variance in 10-year stroke risk beyond age

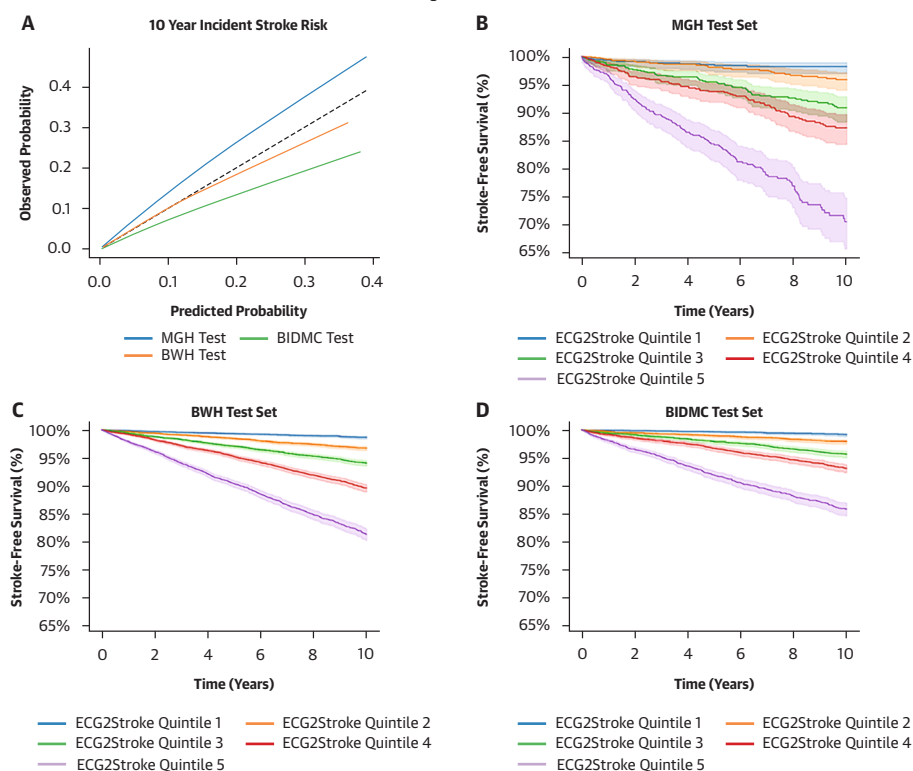
and sex alone (Partial R²: ECG only, 0.082-0.118) (Supplemental Table 5). In both BWH and BIDMC, ECG2Stroke stratified risk across subgroups of sex, age, race, and clinical risk factors while maintaining discrimination and calibration performance (Figure 2, Supplemental Figures 4 and 5, Supplemental Tables 6A and 6B). Decision curves for ECG2Stroke demonstrated net benefit across a range of clinically relevant thresholds (Supplemental Figure 6). Sensitivity, specificity, PPV, and NPV for a range of clinically relevant thresholds are included in Supplemental Table 7. Results were similar when death was modeled as a competing risk, where specifically we observed modest attenuation of effect size (10-year horizon ECG neural network HR: death as censoring, 2.52 [95% CI: 2.44-2.60]; death as competing risk, 2.42 [95% CI: 2.34-2.49]), whereas absolute 10-year stroke risk estimates using the Aalen-Johansen estimator (AJ) were essentially identical to Kaplan-Meier (KM) estimates (MGH test: KM, 0.104, AJ, 0.103; BWH test: KM, 0.072, AJ, 0.071; BIDMC test: KM 0.053, AJ, 0.052).

COMPARISON WITH FSRP. We then compared the performance of ECG2Stroke with the FSRP alone and their combination among individuals with available data. In fitting the model combining both ECG2Stroke and FSRP, both the neural network-predicted probabilities and FSRP were independently associated with incident stroke (neural network probabilities: HR: 2.06 [95% CI: 1.84-2.30]; FSRP: HR: 1.62 [95% CI: 1.37-1.93]), without significant interaction.

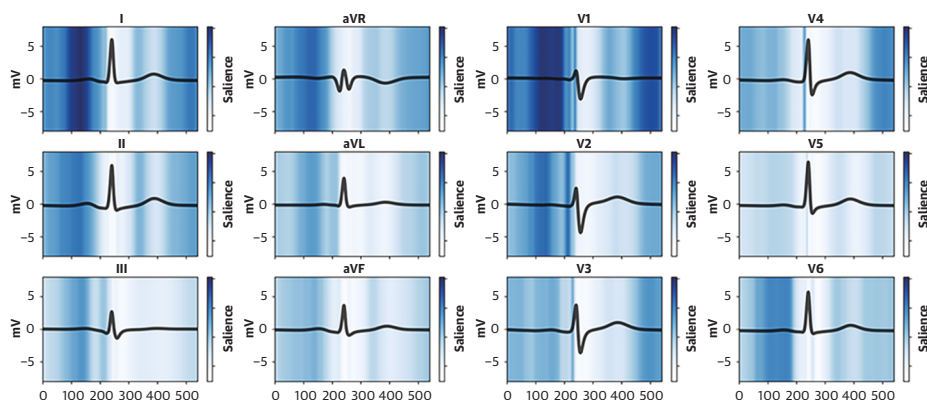
CENTRAL ILLUSTRATION Summary of ECG2Stroke



ECG-Based AI Signatures Predict Stroke Risk



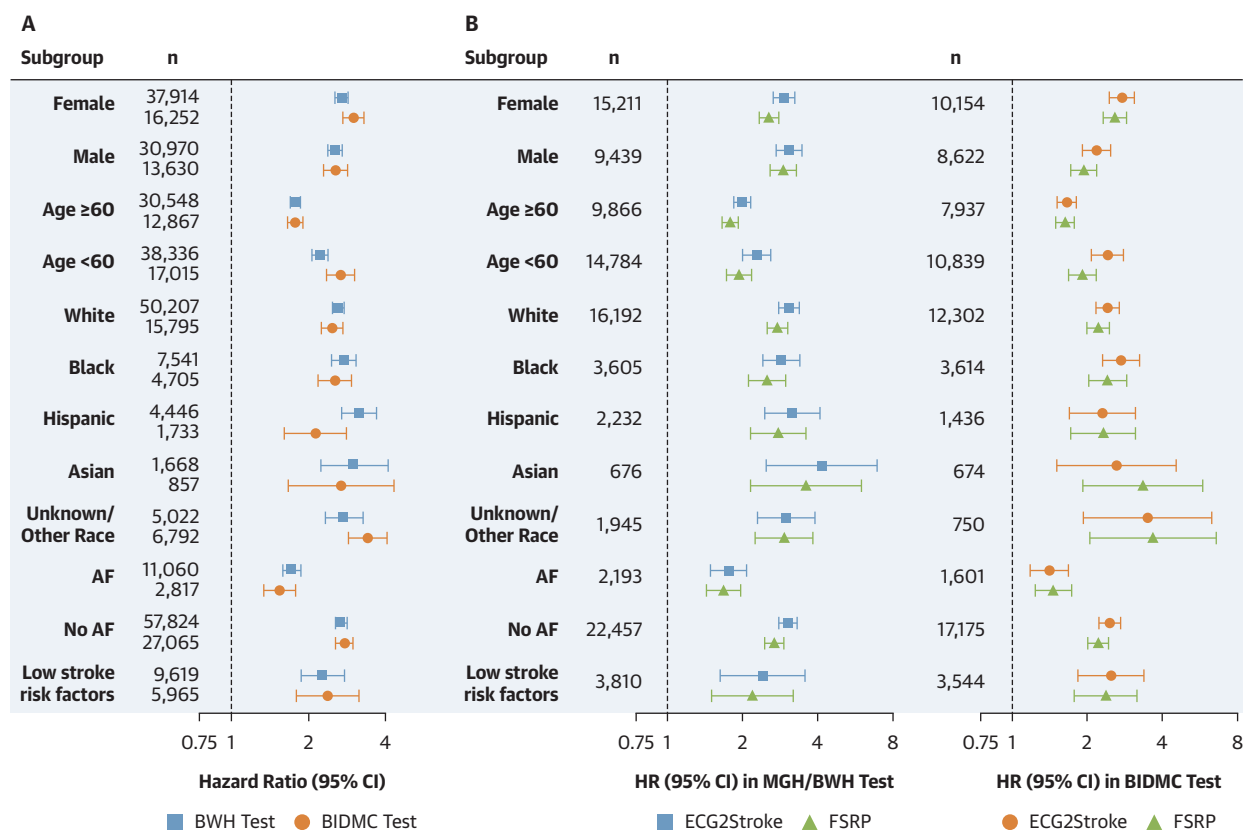
P-Wave and Surrounding Regions Exert the Greatest Influence on Future Risk of Stroke



Mahajan R, et al. JACC. 2026;88(2):205-223.

We developed and validated ECG2Stroke, an artificial intelligence (AI)-enabled model that predicts 10-year risk of ischemic stroke. Calibration and longitudinal stratification of stroke are shown in the middle panels. Saliency mapping (bottom) and correlation analyses demonstrate that ECG2Stroke appears to encode markers of atrial cardiopathy. In the saliency maps, the median waveform for each ECG lead is shown with darker shades of blue denoting areas of the ECG waveform most influential on ECG2Stroke predictions. BIDMC = Beth Israel Deaconess Medical Center; BWH = Brigham and Women's Hospital; ECG = electrocardiogram; MGH = Massachusetts General Hospital.

FIGURE 2 ECG2Stroke Association With Stroke Risk in Clinical and Demographic Subgroups



(A) Forest plot of HRs for 10-year ischemic stroke per 1-SD increase in ECG2Stroke linear predictor for subgroups in both external test sets, BWH (n = 68,884) (blue) and BIDMC (n = 29,882) (orange). The low stroke risk factors subgroup was defined as patients without baseline stroke risk factors including any of: current smoking, diabetes, hyperlipidemia, chronic kidney disease, hypertension, atrial fibrillation, heart failure, valvular disease, vascular disease. (B) HRs for 10-year ischemic stroke per 1-SD increase in linear predictor of either ECG2Stroke or FSRP-based Cox model for subgroups derived from the FSRP-complete subsets of MGH/BWH test (n = 24,650; left axes with blue and green markers) or BIDMC test (n = 18,776; right axes with orange and green markers). AF = atrial fibrillation; FSRP = Framingham Stroke Risk Profile; other abbreviations as in Figure 1.

To ensure sufficient sample size, assessment of FSRP was performed in a combined MGH/BWH test set along with the BIDMC test set (baseline characteristics are presented in Supplemental Table 8). Here, both ECG2Stroke and FSRP demonstrated strong univariable association with 10-year incident stroke in the test sets (MGH/BWH test HR per SD: ECG2Stroke, 3.016 [95% CI: 2.795-3.253]; FSRP, 2.732 [95% CI: 2.534-2.945]; BIDMC test HR per SD: ECG2Stroke, 2.474 [95% CI: 2.269-2.698]; FSRP, 2.301 [95% CI: 2.118-2.500]) (Table 3). The discrimination and calibration of ECG2Stroke at a 10-year horizon were similar to the FSRP-based model but with numerically higher point estimates (MGH/BWH test AUC: ECG2Stroke, 0.791 [95% CI: 0.776-0.806]; FSRP, 0.779 [95% CI: 0.763-0.795]; BIDMC test AUC: ECG2Stroke, 0.745 [95% CI: 0.727-0.763]; FSRP, 0.728

[95% CI: 0.710-0.747]). Discrimination was marginally improved in the model combining ECG2Stroke and FSRP (10-year AUC: MGH/BWH test, 0.800 [95% CI: 0.785-0.815]; BIDMC test, 0.754 [95% CI: 0.735-0.772]) (Table 3). Calibration was favorable in the MGH/BWH test set compared with the BIDMC test set, with similar behavior with the use of ECG2Stroke and FSRP (Table 3, Figure 3). Similar patterns were observed at shorter time horizons (Supplemental Tables 9A and 9B). Both models retained predictive performance across subgroups of age, sex, race, and clinical risk factors in both test sets (Figure 2, Supplemental Tables 10A and 10B). Sensitivity analysis restricting the Cox model training and testing cohorts to age >55 years to match the original FSRP validation cohort resulted in worse FSRP 10-year performance across test sets (AUC: MGH/BWH test,

TABLE 3 Performance of Cox Models for 10-Year Incident Stroke Prediction in Test Set Subsets With Complete FSRP Data

	HR (per SD)	AUC	AP	nAP	ICI
MGH/BWH test set (n = 24,650; event rate = 0.065)					
Age + sex	2.567 (2.373-2.776)	0.753 (0.737-0.770) ^a	0.119 (0.109-0.130) ^a	1.830 (1.710-1.970)	0.007
ECG alone	2.677 (2.497-2.870)	0.771 (0.754-0.788) ^a	0.142 (0.129-0.158)	2.190 (2.030-2.415)	0.008
FSRP	2.732 (2.534-2.945)	0.779 (0.763-0.796)	0.127 (0.117-0.140) ^a	1.960 (1.845-2.120)	0.006
ECG2Stroke	3.016 (2.795-3.253)	0.791 (0.776-0.806)	0.142 (0.129-0.157)	2.190 (2.045-2.390)	0.006
ECG2Stroke and FSRP	3.094 (2.870-3.336)	0.800 (0.787-0.815) ^a	0.145 (0.132-0.160)	2.230 (2.070-2.435)	0.006
BIDMC test set (n = 18,776; event rate = 0.046)					
Age + sex	2.131 (1.957-2.320)	0.709 (0.690-0.728) ^a	0.080 (0.073-0.089) ^a	1.760 (1.630-1.910)	0.035
ECG alone	2.283 (2.106-2.476)	0.729 (0.711-0.746) ^a	0.094 (0.084-0.105)	2.050 (1.890-2.295)	0.037
FSRP	2.301 (2.118-2.500)	0.728 (0.710-0.747) ^a	0.095 (0.085-0.111)	2.080 (1.885-2.400)	0.03
ECG2Stroke	2.474 (2.269-2.698)	0.745 (0.727-0.763)	0.093 (0.084-0.103)	2.030 (1.890-2.250)	0.038
ECG2Stroke and FSRP	2.586 (2.373-2.819)	0.754 (0.735-0.772) ^a	0.101 (0.091-0.116) ^a	2.220 (2.030-2.495)	0.035

Ranges in parentheses are 95% CIs. ^a*P* < 0.05 compared with ECG2Stroke.
FSRP = Framingham Stroke Risk Profile; other abbreviations as in [Tables 1 and 2](#).

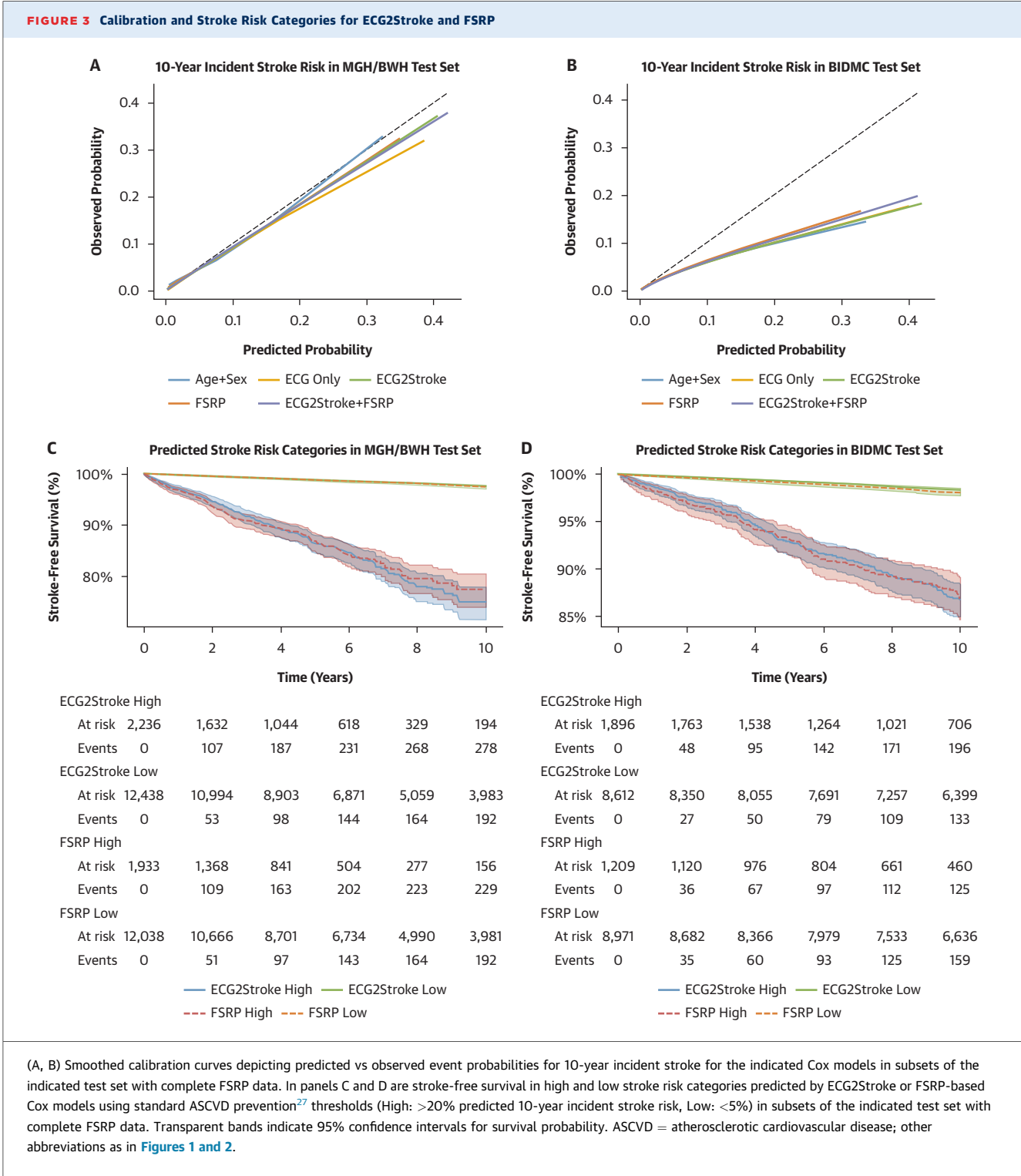
0.737 [95% CI: 0.714-0.757]; BIDMC test, 0.681 [95% CI: 0.660-0.703]; nAP: MGH/BWH test: 1.420 [95% CI: 1.340-1.530]; BIDMC test, 1.590 [95% CI: 1.440-1.810]; ICI: MGH/BWH test: 0.013; BIDMC test, 0.050).

Using low (10-year risk <5%) and high (10-year risk >20%) risk thresholds derived from standard ASCVD prevention guidelines,²⁷ we found similar stratification of longitudinal stroke incidence among ECG2Stroke and FSRP-defined risk groups (MGH/BWH test: low risk, *P* = 0.82; high risk, *P* = 0.86; BIDMC test: low risk, *P* = 0.23; high risk, *P* = 0.97) ([Figure 3](#)). ECG2Stroke improved reclassification modestly at the high vs intermediate risk threshold (MGH/BWH test: NRI overall, 0.048 [95% CI: 0.020-0.075]; NRI+, 0.054 [95% CI: 0.027-0.081]; NRI-, -0.006 [95% CI: -0.010 to 0.002]; BIDMC test: NRI overall, 0.073 [95% CI: 0.044-0.104]; NRI+, 0.092 [95% CI: 0.063-0.123], NRI-, -0.019 [95% CI: -0.024 to -0.015]). Reclassification was not significantly improved at the low vs intermediate risk threshold (MGH/BWH test: NRI overall, 0.002 [95% CI: -0.029 to 0.030]; NRI+, 0.002 [95% CI: -0.026 to 0.028]; NRI-, 0.000 [95% CI: -0.012 to 0.012]; BIDMC test: NRI overall, 0.016 [95% CI: -0.010 to 0.041]; NRI+, 0.034 [95% CI: 0.010-0.057], NRI-, -0.019 [95% CI: -0.026 to -0.011]). Findings were similar with the use of alternate risk thresholds (low 10-year risk <7.5%, high 10-year risk >10%) ([Supplemental Figure 7](#)). Decision curves for 10-year stroke demonstrated modest improvements in net benefit with the use of ECG2Stroke compared with FSRP across a range of relevant thresholds (threshold probabilities: MGH/BWH test, 1%-40%; BIDMC test, 1%-14%)

([Supplemental Figure 8](#)). Sensitivity, specificity, PPV, and NPV, along with net benefit values for a range of clinically relevant thresholds for both models, are included in [Supplemental Table 11](#).

The linear predictors of ECG2Stroke and FSRP-based model were highly correlated in all test sets (Pearson correlation: MGH/BWH test set, 0.82; BIDMC, 0.81; *P* < 0.01 for both). However, variance decomposition confirmed that both ECG2Stroke and FSRP explained a significant proportion of additional variance in 10-year stroke risk beyond the other alone (MGH/BWH partial *R*²: ECG2Stroke, 0.076; FSRP, 0.016; BIDMC partial *R*²: ECG2Stroke, 0.057; FSRP, 0.024) ([Supplemental Table 12](#)).

ATRIAL FIBRILLATION. We examined the effect of AF on model performance owing to its particular relevance to both ischemic stroke risk and the ECG. In the combined MGH/BWH test set, ECG2Stroke discrimination and calibration were more favorable among individuals without vs with prevalent AF ([Table 4](#), [Supplemental Tables 13A and 13B](#), [Supplemental Figure 9](#)). Calibration was again favorable in the MGH/BWH test set compared with the BIDMC test set. Stratification by OAC status in MGH/BWH revealed no major changes in model performance among patients with AF. Similar trends were observed at shorter time horizons. To assess the extent to which ECG2Stroke performance extends beyond prediction of AF, we examined model performance among patients without AF at baseline with censoring at incident AF diagnosis, observing continued moderate discrimination and good calibration ([Table 4](#), [Supplemental Tables 13A and 13B](#)).

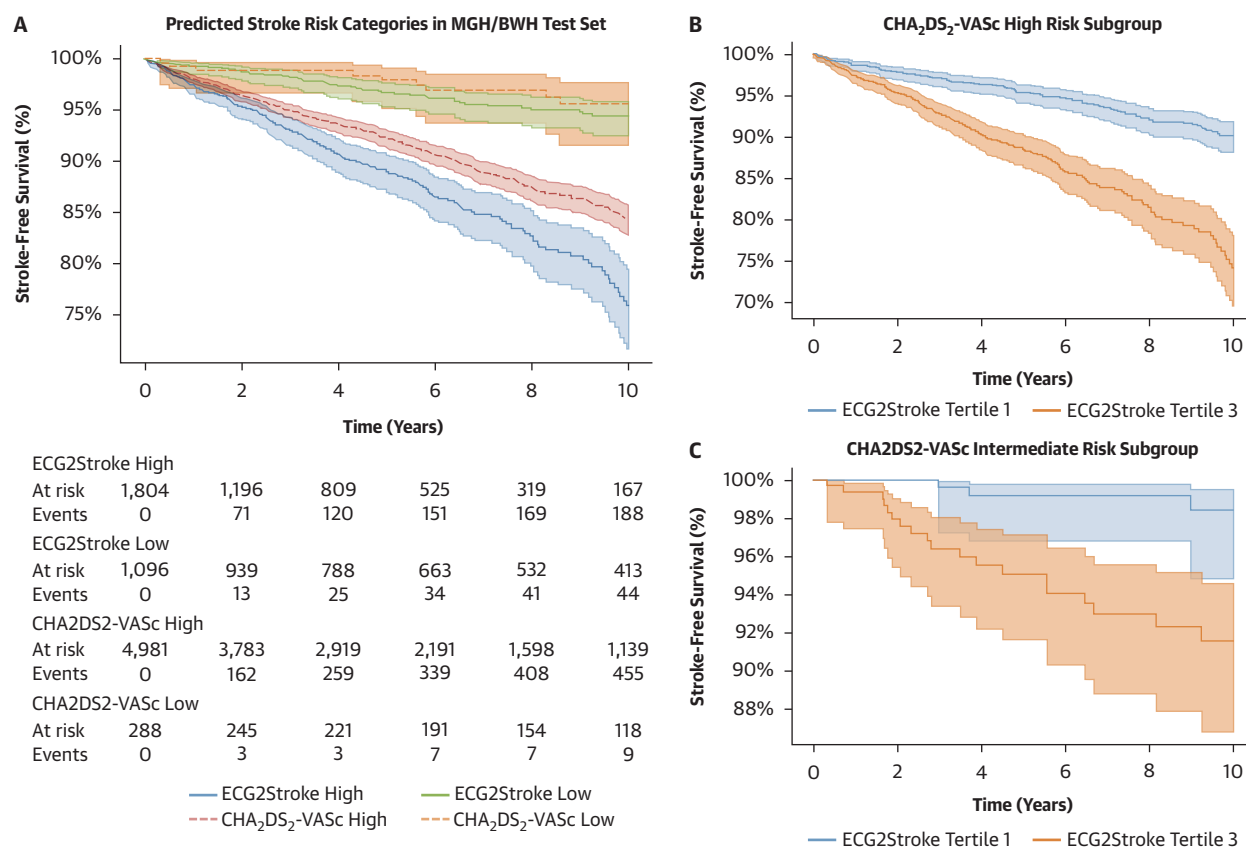


In an exploratory analysis, we compared classification of AF patients not on OAC at baseline into low and high stroke risk groups according to standard risk thresholds for CHA₂DS₂-VAsC scores (low, <1 for men and <2 for women; intermediate, 1 for men and 2 for women; high, >1 for men and >2 for women) and standard ASCVD prevention risk thresholds (10-year risk <5% for low, >20% for high) for ECG2Stroke in

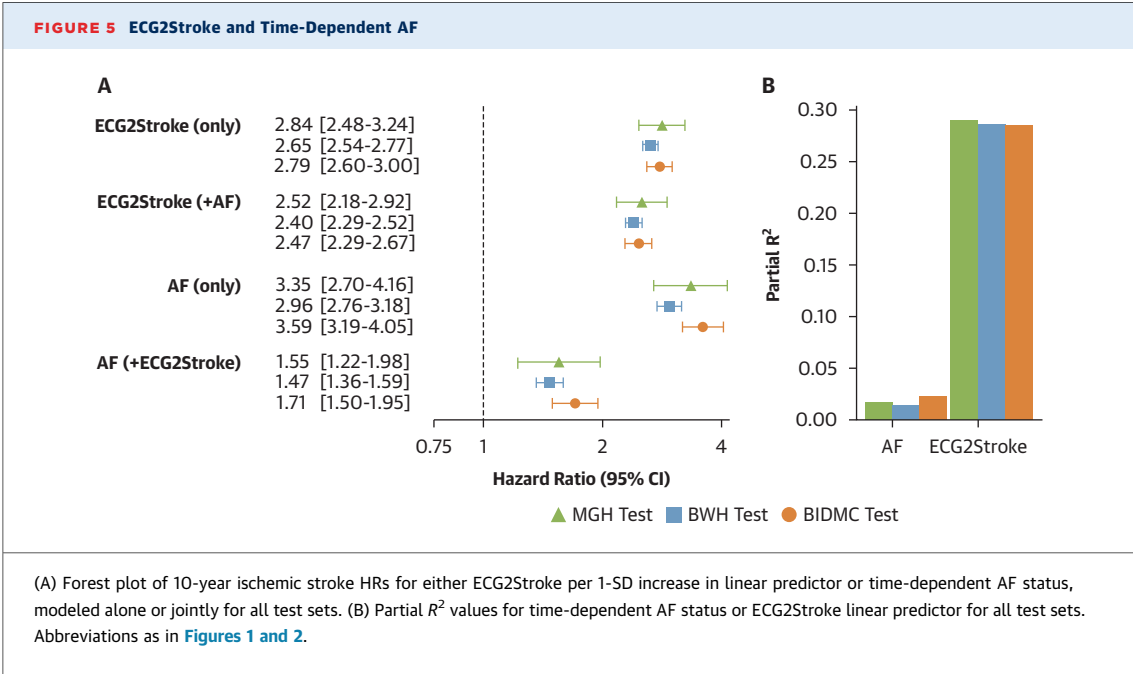
TABLE 4 ECG2Stroke Performance for 10-Year Incident Stroke Prediction in AF-Related Subgroups

	n	HR (per SD)	AUC	AP	Event Rate	nAP	ICI
MGH/BWH test set subgroups							
No AF	61,880	2.695 (2.568-2.828)	0.764 (0.754-0.775)	0.127 (0.120-0.135)	0.064	1.990 (1.880-2.130)	0.001
No AF, censored at AF	61,880	2.612 (2.477-2.753)	0.764 (0.752-0.774)	0.099 (0.093-0.106)	0.055	1.810 (1.700-1.930)	0.007
AF	11,775	1.760 (1.626-1.905)	0.716 (0.697-0.737)	0.161 (0.149-0.175)	0.142	1.140 (1.050-1.240)	0.029
AF –OAC	6,228	1.899 (1.697-2.125)	0.724 (0.699-0.751)	0.156 (0.141-0.176)	0.132	1.180 (1.060-1.330)	0.026
AF +OAC	5,547	1.585 (1.420-1.769)	0.688 (0.654-0.723)	0.171 (0.152-0.196)	0.155	1.100 (0.985-1.270)	0.033
BIDMC test set subgroups							
No AF	27,065	2.769 (2.554-3.003)	0.758 (0.743-0.774)	0.094 (0.086-0.105)	0.046	2.050 (1.880-2.290)	0.024
No AF, censored at AF	27,065	2.682 (2.458-2.926)	0.759 (0.743-0.776)	0.075 (0.068-0.083)	0.041	1.820 (1.650-2.020)	0.027
AF	2,817	1.543 (1.338-1.779)	0.726 (0.692-0.759)	0.141 (0.123-0.164)	0.13	1.080 (0.945-1.260)	0.059

Ranges in parentheses are 95% CIs.
AF = atrial fibrillation; OAC = oral anticoagulation; other abbreviations as in Table 1 and 2.

FIGURE 4 ECG2Stroke and CHA₂DS₂-VASC Stroke Risk Categories in Patients With AF

(A) Stroke-free survival in high and low stroke risk categories predicted by ECG2Stroke (high: >20% predicted 10-year incident stroke risk; low: <5%) or CHA₂DS₂-VASC score (high: >1 for men and >2 for women; low: <1 for men and <2 for women) in MGH/BWH test set patients with AF not on OAC at baseline. (B, C) Risk stratification by tertiles of ECG2Stroke within CHA₂DS₂-VASC score-defined (B) high risk category (>1 for men and >2 for women) or (C) intermediate risk category (1 for men and 2 for women) for MGH/BWH test set patients with AF not on OAC at baseline. Shaded bands indicate 95% CIs for survival probability in all panels. OAC = oral anticoagulation; other abbreviations as in Figures 1 and 2.



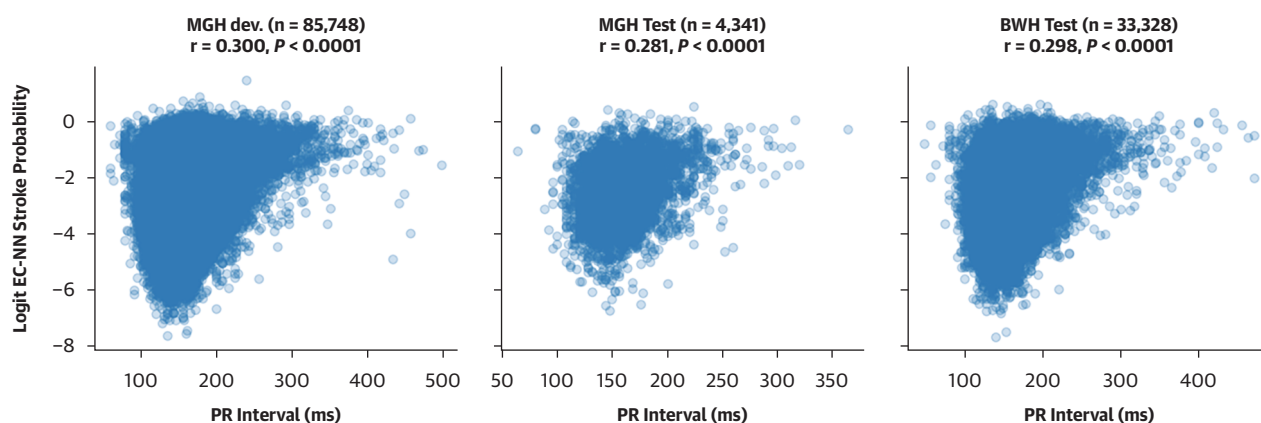
the MGH/BWH test set ([Figure 4](#)). Stroke-free survival differed between the high risk groups (log-rank $P < 0.005$), but not the low risk groups ($P = 0.43$). The ECG2Stroke low risk group included more patients ($n = 1,096$) than the corresponding CHA₂DS₂-VASc group ($n = 288$), whereas the reverse was true for high risk groups (ECG2Stroke, $n = 1,804$; CHA₂DS₂-VASc, $n = 4,981$). Quantiles of ECG2Stroke were also able to stratify risk further within CHA₂DS₂-VASc-defined high risk and intermediate risk groups ([Figure 4](#)). Results were similar if patients were censored at OAC initiation ([Supplemental Figure 10](#)) or with use of alternative risk thresholds ([Supplemental Figure 11](#)). ECG2Stroke applied at standard risk thresholds improved reclassification at the low vs intermediate risk threshold (NRI overall, 0.119 [95% CI: 0.082-0.155]; NRI+, -0.068 [95% CI: -0.098 to -0.043]; NRI-, 0.187 [95% CI: 0.165-0.210]) and at the high vs intermediate threshold (NRI overall, 0.071 [95% CI: 0.018-0.126]; NRI+, -0.546 [95% CI: -0.590 to -0.495]; NRI-, 0.616 [95% CI: 0.593-0.640]), driven primarily by appropriate down-classification of nonevents. Use of the less stringent alternative risk thresholds (10-year risk $<7.5\%$ for low, $>10\%$ for high) for ECG2Stroke further increased reclassification improvement (low vs intermediate threshold NRI: overall, 0.206 [95% CI: 0.161-0.248]; NRI+, -0.141 [95% CI: -0.177 to -0.107]; (NRI-, 0.347 [95% CI: 0.322-0.372]; high vs intermediate threshold NRI: overall, 0.117 [95% CI:

0.069-0.166]; NRI+, -0.162 [95% CI: -0.201 to -0.121]; NRI-, 0.279 [95% CI: 0.252-0.304]). Decision curves for 10-year stroke demonstrated substantial improvements in net benefit with the use of ECG2Stroke compared with CHA₂DS₂-VASc across a range of clinically relevant thresholds (threshold probabilities: MGH/BWH test, 1%-24%) ([Supplemental Figure 12](#)). Sensitivity, specificity, PPV, and NPV, along with net benefit values for a range of clinically relevant thresholds for both models, are included in [Supplemental Table 14](#).

Time-dependent modeling of AF status to directly characterize relations between ECG2Stroke, AF, and 10-year stroke risk revealed greater coefficient attenuation with joint modeling for AF than ECG2Stroke, much larger partial R² for ECG2Stroke, and a moderate submultiplicative interaction between the two (HR for interaction range: 0.658-0.745) ([Figure 5](#), [Supplemental Tables 15A and 15B](#)). Results were similar when we accounted for potential occult subclinical AF by treating AF detected up to 2 years after stroke as prevalent AF ([Supplemental Tables 15A and 15B](#)).

CLINICAL AND ECG CORRELATES OF ECG-BASED STROKE RISK. Examination of ECG neural network saliency maps demonstrated that variation in the ECG waveform near the region of the P-wave had the greatest effect on model predictions ([Central Illustration](#)). We examined associations of the ECG

FIGURE 6 PR-Interval Correlations With ECG Neural Network Output



Depicted are correlation plots between ECG PR interval (x-axis) and logit-transformed ECG neural network probability (y-axis). Abbreviations as in [Figure 1](#).

neural network component of ECG2Stroke with available ECG-based atrial substrate markers and found moderate correlation with PR interval (Pearson R : MGH development, 0.300; MGH test, 0.281; BWH test, 0.298; $P < 0.001$ for all) ([Figure 6](#)) and modest but significant correlations with P-wave duration (Pearson R : MGH test, 0.039; BWH test, 0.033; $P < 0.001$ for all) and P-wave axis (Pearson R : MGH test, 0.034; BWH test, 0.044; $P < 0.001$ for all) ([Supplemental Figure 13](#)). A Cox model including all 3 P-wave ECG metrics (PR interval, P-wave duration, P-axis) as predictors trained in the MGH development set retained some discriminative performance for stroke in the test sets but was significantly worse than models based on ECG or age + sex ([Supplemental Table 16](#)).

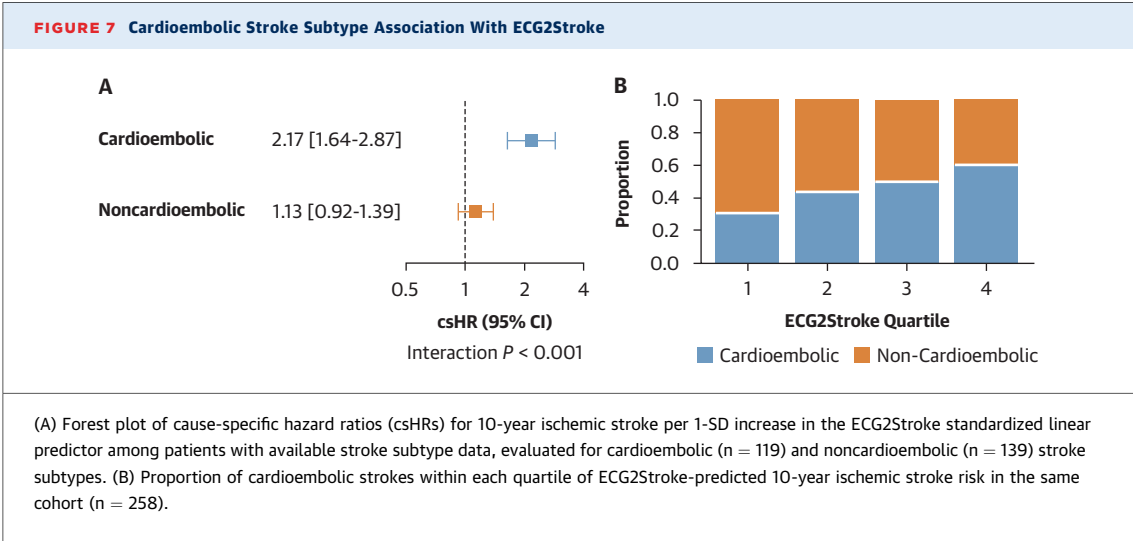
To quantify model specificity, we examined associations between stroke risk models and alternative incident cardiovascular outcomes including HF, MI, and death. Here, although both ECG2Stroke and FSRP discriminated each of the alternative outcomes ([Supplemental Table 17](#)), there was substantially better agreement between observed and expected probabilities (ie, calibration) for incident stroke compared with HF and death, although generally similar for MI ([Supplemental Figure 14](#)). We further quantified associations with CE vs non-CE stroke in a small subset of patients with incident strokes that were manually subtyped (CE, $n = 119$; non-CE, $n = 139$). ECG2Stroke-predicted 10-year stroke risk was strongly associated with CE stroke but not with non-CE stroke (cause-specific HR per 1-SD increase in linear predictor: CE, 2.168 [95% CI: 1.641-2.865]; non-

CE, 1.133 [95% CI: 0.923-1.391]; P value for heterogeneity: <0.001) ([Figure 7](#)). Consistently with this, patients with ECG2Stroke-predicted 10-year stroke risk below the median had disproportionately more non-CE strokes (top 50% predicted stroke risk: CE, 71/129 [55%], bottom 50% predicted stroke risk: CE, 48/129 [37%]; chi-square = 7.549; $P < 0.01$). Among baseline clinical factors, ECG2Stroke demonstrated strongest associations with HF, AF, and chronic kidney disease ([Figure 8](#)).

DISCUSSION

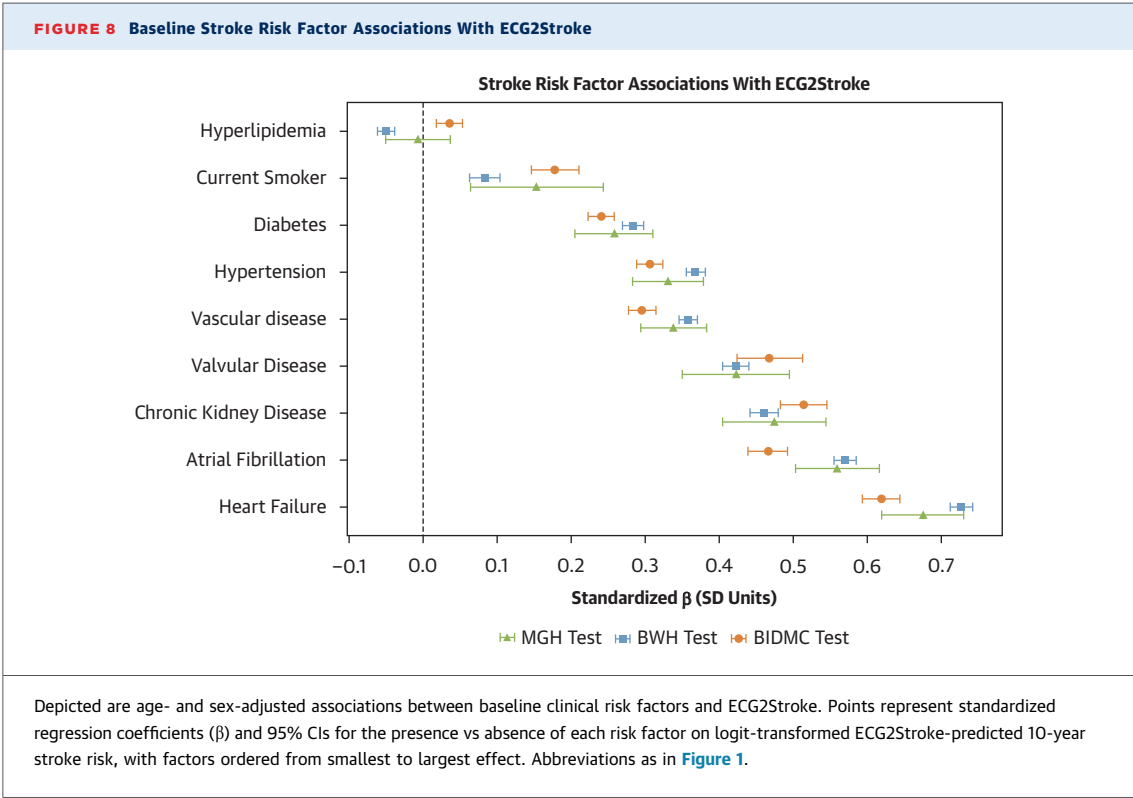
We developed ECG2Stroke, a model to explicitly predict time to incident ischemic stroke, by survival-aware deep learning-enabled AI analysis of raw 12-lead ECG waveforms, as trained on >300,000 ECGs from >100,000 individuals within established ambulatory care cohorts. ECG2Stroke demonstrated consistent predictive utility across 3 test sets, including 2 large independent datasets spanning clinically and demographically varied populations. Among individuals with available clinical data, ECG2Stroke provided similar discrimination of stroke risk compared with the validated FSRP clinical risk model. ECG2Stroke showed strong associations with clinical and ECG correlates of atrial dysfunction as well as CE stroke, pointing to plausible risk mechanisms related to atrial cardiopathy ([Central Illustration](#)).

Our work extends the growing literature on AI-enabled ECG-based risk assessment, which to date has focused largely on cardiac phenotypes or broad



cardiovascular outcomes.^{10,12-14,34-37} Hughes et al³⁷ trained an AI-enabled ECG-based risk score for 5-year cardiovascular death and composite atherosclerotic outcomes, demonstrating incremental value when combined with the Pooled Cohorts Equations; however, stroke was included only as part of the composite outcome. Lin et al³⁶ reported a multitask

model for 1-year major adverse cardiovascular events, including ischemic stroke, in a Taiwanese health system, demonstrating feasibility of short-term risk prediction but with limited stroke-specific external validation. Although assessment of broad composite outcomes may facilitate favorable statistical power,^{38,39} actionable insights for clinical



decision making often require study of specific diseases, including dedicated evaluation of disease-specific risk factors and treatment decisions.^{5,6,30}

Our findings demonstrate the potential value of a stroke-specific ECG-based AI model. Croon et al examined phenotypic selectivity of AI-enabled ECG models among multiple cardiovascular phenotypes and found strong evidence for high overlap and low specificity; however, the study focused on intracardiac left ventricular and left-sided valvular phenotypes and did not include broader disease entities or extracardiac disease, such as stroke.¹⁵ Furthermore, the extent to which such overlap in prognostic performance reflects shared underlying risk and pathophysiology vs off-target noise remains unknown.

In the present study, we found that although ECG2Stroke and FSRP also discriminate risk of non-stroke cardiovascular outcomes, agreement between observed and expected probabilities (ie, calibration) is substantially better for stroke than for other cardiovascular outcomes, such as HF and death, supporting a degree of outcome specificity. Of note, the phenomenon of capturing elements of global cardiovascular risk rather than strictly isolating a particular phenotype from other cardiovascular outcomes is not unique to AI-enabled ECG models. In our data, the clinical FSRP score also discriminated nonstroke outcomes, a pattern which has also been previously described for a number of other clinical risk models, including CHA₂DS₂-VASc (AF-related stroke) and CHARGE-AF (incident AF) scores,⁴⁰ CHA₂DS₂-VASc (AF-related stroke) and non-AF mortality,⁴¹ PREVENT equations (ASCVD) and all-cause mortality,⁴² and the GRACE score (ACS-related mortality) and ACS-related stroke.⁴³ On balance, because guidelines appropriately rely on the importance of absolute risk estimates to guide clinical decisions, our findings reinforce the continued relevance of stroke-specific risk estimation in the era of ECG-based AI analysis.

Our findings demonstrate that ECG2Stroke possesses value for stroke risk estimation regardless of the presence or absence of AF. AF is a leading cause of CE stroke, and its presence is detectable using AI-based ECG analysis even when AF is not present on the ECG. Our results reflect the expected association between AF and stroke risk, and our saliency maps (ie, substantial attention near the ECG P-wave) share some analogy to attention maps for AF-related models.¹² However, our findings clarify that the value of ECG2Stroke extends beyond correlation with AF risk, as ECG2Stroke maintained strong association with stroke risk and partial R^2 values after adjusting for time-dependent AF, and continued to

discriminate stroke risk among patients with known AF or when considering all person-time free of AF. Although not specifically trained for AF-related stroke, ECG2Stroke provided information complementary to the guideline-recommended AF-related stroke risk categories defined by the CHA₂DS₂-VASc score,^{4,30} including refinement of intracategory risk and favorable net reclassification, especially for identifying low risk patients. Although the potential to identify additional AF patients at lower stroke risk than expected whose risk-benefit balance may actually favor avoiding anticoagulation is potentially of significant impact, we note that our AF analyses were performed in a smaller subset and require additional external and prospective validation.

Our examination of the clinical and ECG correlates of ECG2Stroke provides conceptual support for the ability of ECG-based AI to encode risk signals specific to CE stroke (eg, atrial cardiopathy). Prominent weighting of the ECG P-wave in our saliency maps and significant correlations to structured P-wave indices consistently implicate the relevance of abnormal atrial substrate to the ECG2Stroke signal. Furthermore, in a small sample of manually subtyped strokes, our finding of a strong association with CE stroke, without a corresponding association with non-CE stroke, further strengthens the plausibility of this proposed mechanism. Rather than trivially predicting AF, our model's performance across AF strata and persistent association with stroke after adjustment for AF with a moderate submultiplicative interaction provides support for ECG2Stroke's potential ability to encode atrial electrical and mechanical dysfunction broadly and in excess of a simple marker of AF risk, consistent with conceptual models of atrial cardiopathy.⁴⁴ In summary, we envision that our approach may provide an ECG biomarker of atrial substrate enabling future mechanistic and interventional studies, and to support such avenues our model has been made publicly available under an academic use license.

Overall, our findings highlight the potential of ECG-based AI to enable scalable stroke risk stratification. Although all patients may benefit from lifestyle choices to promote cerebrovascular health,⁴⁵ long-term stroke risk estimation offers the potential to efficiently prioritize individuals for intensification of general cardiovascular and stroke-specific primary prevention efforts, including improved control of known ASCVD risk factors² or consideration of rhythm monitoring. Our approach supports the concept of stroke risk estimation at scale by relying only on a single noninvasive, relatively low-cost, and ubiquitous modality: the 12-lead ECG. While

ECG2Stroke was strongly correlated with the FSRP, suggesting that the model encodes established clinical risk factor information, we also observed clear evidence of complementarity, including significant partial R^2 values, independent associations with incident stroke when jointly modeled, and additive improvements in discrimination when combined. Furthermore, ECG2Stroke retained predictive performance in patient subgroups without significant clinical risk factors for stroke. In aggregate, our findings suggest that optimal stroke risk estimation may be context dependent, such that consideration of both clinical factors and AI-based ECG analysis may be preferable when both are available, yet ECG-based AI alone can provide moderate risk discrimination in settings where clinical data are not easily at hand or may be subject to misclassification (eg, resource-constrained settings without robust EHR platforms). We acknowledge that although our findings support the feasibility of stroke estimation using ECG-based AI, the present study does not establish clinical utility, and future prospective implementation studies are needed before integration into real-world clinical workflows is justified.

STUDY LIMITATIONS. Limitations of the present study include its retrospective observational design, warranting prospective validation and pragmatic implementation in real-world settings. Although findings were validated in multiple distinct health care samples, all of the institutions are located in a single geographic region, so future validation in more geographically and demographically diverse populations is warranted. Covariates and outcomes were ascertained from EHRs and are subject to potential misclassification, although we used published and validated disease definitions wherever possible.^{19,20} Misclassification of prevalent disease as incident cannot fully be eliminated, although our cohort designs instituted a prolonged ascertainment period before start of follow-up, and we implemented a 30-day blanking period in the present study.^{16,17} ECGs required for study inclusion were ascertained for clinical reasons and therefore introduce selection bias. However, previous work demonstrates that our analysis of cohorts selected on the basis of longitudinal primary or cardiology care, rather than simply data availability, reduces the effects of selection bias.^{16,17} Nevertheless, future validation in populations with systematic ECG assessments would be useful. Given sample size constraints and the exploratory nature of the analysis, our small subset with available stroke subtype adjudications included individuals from the ECG2Stroke training set;

however, we think that this is unlikely to have biased the subtype analysis, because stroke subtype labels were not included as targets in model training. We focused on the ECG because it is well suited for population-based assessment owing to its cost and potential scalability. Future models should assess the value of additional unstructured data (eg, imaging), although their role in stroke risk stratification would be more likely opportunistic rather than population based.

CONCLUSIONS

Requiring only a single 12-lead ECG along with age and sex, ECG2Stroke achieved similar prediction of 10-year incident stroke compared with a validated 8-component clinical risk model across multiple health care systems and within several important clinical subgroups, possibly by encoding signals of abnormal atrial substrate. Although prospective validation and implementation studies are needed, ECG2Stroke may enable prioritization of high-risk patients for intensive primary prevention of stroke and facilitate future mechanistic studies.

DATA AVAILABILITY The ECG2Stroke AI model and weights are available under an academic use license at https://github.com/broadinstitute/ml4h/tree/master/model_zoo/ECG2Stroke.

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APPENDIX For supplemental methods, tables, and figures, please see the online version of this paper.